Certainty Thermometer Classroom Run Kit

TOK Cognitive Science Activity Bank • Cross-AOK • 25-35 min

Category	Cross-AOK	**Time**	25-35 min
Difficulty	Easy	**ID**	certainty-thermometer

Big idea: Certainty is not one simple measure; it depends on evidence, method, context, and purpose.

One-Minute Teacher Brief

Students place claims from different AOKs on a certainty scale, then revise the scale after adding criteria.

Student Prompt

Inspect Certainty Thermometer Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Certainty Thermometer Stimulus A	Student-facing stimulus 1: Claim cards: '2+2=4', 'This source is reliable', 'This song is sad', 'The medicine works', 'The policy is fair'. Name the AOK, the method being used, and the standard of evidence you would apply.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Certainty Thermometer Stimulus B	Student-facing stimulus 2: Criteria cards: proof, replication, consensus, interpretation, usefulness, ethical stakes. Name the AOK, the method being used, and the standard of evidence you would apply.	Use this for comparison, a second round, or a partner challenge.
Certainty Thermometer Reveal / Twist	Student-facing stimulus 3: Criteria cards: proof, replication, consensus, interpretation, usefulness, ethical stakes. Name the AOK, the method being used, and the standard of evidence you would apply.	Teacher reveal: connect the changed response to the big idea — Certainty is not one simple measure; it depends on evidence, method, context, and purpose.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Certainty Thermometer Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Certainty Thermometer Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	Is certainty the best measure of knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Cross-AOK.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how certainty is not one simple measure; it depends on evidence, method, context, and purpose.

• Strong response: separates observation from inference.

• Strong response: names a method, source, model, language choice, context, or perspective.

• Strong response: revises confidence when new evidence or a new criterion appears.

• Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

• Work silently for the first response so you can notice your own starting point.

• Record one knowledge claim, the evidence behind it, and your confidence level.

• After the reveal or comparison, update your claim in a different colour or column.

• Use the debrief to answer: Is certainty the best measure of knowledge?

Setup Checklist

• Make the certainty scale physical if possible, using the wall or floor.

• Start with intuitive placement before introducing criteria cards.

Example Stimuli or Materials

• Claim cards: '2+2=4', 'This source is reliable', 'This song is sad', 'The medicine works', 'The policy is fair'.

• Criteria cards: proof, replication, consensus, interpretation, usefulness, ethical stakes.

Resources To Use

• Resource pack: resources/certainty-thermometer/index.html

• Card deck CSV: resources/certainty-thermometer/cards.csv

• Visual prompt: resources/certainty-thermometer/visual-prompt.svg

• Certainty thermometer poster from low confidence to high certainty.

• Revision log: original placement, criterion added, revised placement, reason.

Run Sequence

1. Give groups claim cards and ask them to place each on a certainty thermometer.
2. Groups explain the criterion they used first.
3. Introduce criteria cards such as proof, replication, interpretation, consensus, usefulness, and ethical stakes.
4. Students revise placements using at least two criteria.
5. Debrief why certainty differs across knowledge contexts.

Debrief Moves

• Knowledge question: Is certainty the best measure of knowledge?

• Can less certain knowledge be more useful or meaningful?

• How should knowers compare standards across AOKs?

Teacher Quick Script

• Certainty is not the only value knowledge can have.

• Which claim moved when you changed the criterion?

Exit Ticket

• One thing I first treated as evidence was...

• My confidence changed because...

• A better knowledge question I can now ask is...